

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import lit
```

```
spark = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()
print("Spark Version:", spark.version)
```

Spark Version: 4.0.2

```
a = spark.read.csv("world.csv", header=True, inferSchema=True)
a
```

```
DataFrame[RoundID: int, MatchID: int, TeamInitials: string, CoachName: string, Line-up: string, ShirtNumber: int,
PlayerName: string, Position: string, Event: string]
```

```
a.printSchema()
```

```
root
|-- RoundID: integer (nullable = true)
|-- MatchID: integer (nullable = true)
|-- TeamInitials: string (nullable = true)
|-- CoachName: string (nullable = true)
|-- Line-up: string (nullable = true)
|-- ShirtNumber: integer (nullable = true)
|-- PlayerName: string (nullable = true)
|-- Position: string (nullable = true)
|-- Event: string (nullable = true)
```

```
row_count = a.count()
print("Number of rows: {row_count}")
```

```
column_count = len(a.columns)
print("Number of columns: {column_count}")
```

```
Number of rows: {row_count}
Number of columns: {column_count}
```

```
a.select("PlayerName", "CoachName").show()
```

```
+-----+-----+
|      PlayerName|      CoachName|
+-----+-----+
|      Alex THEPOT|CAUDRON Raoul (FRA)|
| Oscar BONFIGLIO|  LUQUE Juan (MEX)|
| Marcel LANGILLER|CAUDRON Raoul (FRA)|
| Juan CARRENO|  LUQUE Juan (MEX)|
| Ernest LIBERATI|CAUDRON Raoul (FRA)|
| Rafael GARZA|  LUQUE Juan (MEX)|
| Andre MASCHINOT|CAUDRON Raoul (FRA)|
| Hilario LOPEZ|  LUQUE Juan (MEX)|
| Etienne MATTLER|CAUDRON Raoul (FRA)|
| Dionisio MEJIA|  LUQUE Juan (MEX)|
| Marcel PINEL|CAUDRON Raoul (FRA)|
| Felipe ROSAS|  LUQUE Juan (MEX)|
| Alex VILLAPLANE|CAUDRON Raoul (FRA)|
| Manuel ROSAS|  LUQUE Juan (MEX)|
| Lucien LAURENT|CAUDRON Raoul (FRA)|
| Jose RUIZ|  LUQUE Juan (MEX)|
| Marcel CAPELLE|CAUDRON Raoul (FRA)|
| Alfredo SANCHEZ|  LUQUE Juan (MEX)|
| Augustin CHANTREL|CAUDRON Raoul (FRA)|
| Efrain AMEZCUA|  LUQUE Juan (MEX)|
+-----+-----+
only showing top 20 rows
```

```
filtered_df = a.filter(a.MatchID == 1096)
filtered_df.show()
print("Number of records with MatchID = 1096: {filtered_df.count()}")
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|RoundID|MatchID|TeamInitials|      CoachName|Line-up|ShirtNumber|      PlayerName|Position|      Event|
+-----+-----+-----+-----+-----+-----+-----+-----+
|      201|      1096|      FRA|CAUDRON Raoul (FRA)|      S|      0|      Alex THEPOT|      GK|      NULL|
|      201|      1096|      MEX|  LUQUE Juan (MEX)|      S|      0| Oscar BONFIGLIO|      GK|      NULL|
|      201|      1096|      FRA|CAUDRON Raoul (FRA)|      S|      0| Marcel LANGILLER|      NULL|      G40'|
|      201|      1096|      MEX|  LUQUE Juan (MEX)|      S|      0| Juan CARRENO|      NULL|      G70'|
|      201|      1096|      FRA|CAUDRON Raoul (FRA)|      S|      0| Ernest LIBERATI|      NULL|      NULL|
|      201|      1096|      MEX|  LUQUE Juan (MEX)|      S|      0| Rafael GARZA|      C|      NULL|
|      201|      1096|      FRA|CAUDRON Raoul (FRA)|      S|      0| Andre MASCHINOT|      NULL|G43' G87'|
```

201	1096	MEX	LUQUE Juan (MEX)	S	0	Hilario LOPEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Etienne MATTIER	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Dionisio MEJIA	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel PINEL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Felipe ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Alex VILLAPLANE	C	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Manuel ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Lucien LAURENT	NULL	G19'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Jose RUIZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel CAPELLE	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Alfredo SANCHEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Augustin CHANTREL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Efrain AMEZCUA	NULL	NULL

only showing top 20 rows

Number of records with MatchID = 1096: {filtered_df.count()}

a.orderBy("MatchID").show()

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
323	25	BRA	LAZARONI Sebastia...	S	1	TAFFAREL	GK	NULL
323	25	BRA	LAZARONI Sebastia...	S	21	MAURO GALVAO	NULL	Y50' 083'
323	25	ARG	BILARDO Carlos (ARG)	S	12	Sergio GOYCOCHEA	GK	Y87'
323	25	BRA	LAZARONI Sebastia...	S	2	JORGINHO	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	4	Jose BASUALDO	NULL	NULL
323	25	BRA	LAZARONI Sebastia...	S	3	RICARDO GOMES	C	R85'
323	25	ARG	BILARDO Carlos (ARG)	S	7	Jorge BURRUCHAGA	NULL	NULL
323	25	BRA	LAZARONI Sebastia...	S	4	DUNGA	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	8	Claudio CANIGGIA	NULL	G81'
323	25	BRA	LAZARONI Sebastia...	S	5	ALEMAO	NULL	083'
323	25	ARG	BILARDO Carlos (ARG)	S	10	Diego MARADONA	C	NULL
323	25	BRA	LAZARONI Sebastia...	S	6	BRANCO	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	14	Ricardo GIUSTI	NULL	Y28'
323	25	BRA	LAZARONI Sebastia...	S	8	VALDO	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	15	Pedro MONZON	NULL	Y27'
323	25	BRA	LAZARONI Sebastia...	S	9	CARECA	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	16	Julio OLARTICOECHEA	NULL	NULL
323	25	BRA	LAZARONI Sebastia...	S	15	MULLER	NULL	NULL
323	25	ARG	BILARDO Carlos (ARG)	S	19	Oscar RUGGERI	NULL	NULL
323	25	BRA	LAZARONI Sebastia...	S	19	RICARDO ROCHA	NULL	Y40'

only showing top 20 rows

a.filter((a.Position == 'C') & (a.Event == 'G40')).show()

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
201	1089	PAR	DURAND LAGUNA Jos...	S	0	Luis VARGAS PENA	C	G40'
429	1175	HUN	DIETZ Karoly (HUN)	S	0	Gyorgy SAROSI	C	G40'

a.select("MatchID").distinct().show()

MatchID
1088
1238
833
463
43950027
1084
1507
1460
1990
392
243
43950022
300061467
300186473
300186487
1143
1339
1650
31
97410013

only showing top 20 rows

a.createOrReplaceTempView("world_data")

spark.sql("SELECT PlayerName, CoachName FROM world_data").show()

```

+-----+-----+
|      playerName      |      CoachName      |
+-----+-----+
| Alex THEPOT | CAUDRON Raoul (FRA) |
| Oscar BONFIGLIO | LUQUE Juan (MEX) |
| Marcel LANGILLER | CAUDRON Raoul (FRA) |
| Juan CARRENO | LUQUE Juan (MEX) |
| Ernest LIBERATI | CAUDRON Raoul (FRA) |
| Rafael GARZA | LUQUE Juan (MEX) |
| Andre MASCHINOT | CAUDRON Raoul (FRA) |
| Hilario LOPEZ | LUQUE Juan (MEX) |
| Etienne MATTTLER | CAUDRON Raoul (FRA) |
| Dionisio MEJIA | LUQUE Juan (MEX) |
| Marcel PINEL | CAUDRON Raoul (FRA) |
| Felipe ROSAS | LUQUE Juan (MEX) |
| Alex VILLAPLANE | CAUDRON Raoul (FRA) |
| Manuel ROSAS | LUQUE Juan (MEX) |
| Lucien LAURENT | CAUDRON Raoul (FRA) |
| Jose RUIZ | LUQUE Juan (MEX) |
| Marcel CAPELLE | CAUDRON Raoul (FRA) |
| Alfredo SANCHEZ | LUQUE Juan (MEX) |
| Augustin CHANTREL | CAUDRON Raoul (FRA) |
| Efrain AMEZCUA | LUQUE Juan (MEX) |
+-----+-----+
only showing top 20 rows

```

```
a.withColumn("year", lit("1930")).show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|RoundID|MatchID|TeamInitials|      CoachName      |Line-up|ShirtNumber|      PlayerName      |Position|      Event      |year|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Alex THEPOT | GK | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Oscar BONFIGLIO | GK | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Marcel LANGILLER | NULL | G40' | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Juan CARRENO | NULL | G70' | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Ernest LIBERATI | NULL | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Rafael GARZA | C | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Andre MASCHINOT | NULL | G43' G87' | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Hilario LOPEZ | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Etienne MATTTLER | NULL | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Dionisio MEJIA | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Marcel PINEL | NULL | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Felipe ROSAS | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Alex VILLAPLANE | C | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Manuel ROSAS | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Lucien LAURENT | NULL | G19' | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Jose RUIZ | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Marcel CAPELLE | NULL | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Alfredo SANCHEZ | NULL | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Augustin CHANTREL | NULL | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Efrain AMEZCUA | NULL | NULL | 1930 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```
from pyspark.sql.functions import desc,avg,col,max,min,sum,count
```

```
a = spark.read.csv("mtcars.csv", header=True, inferSchema=True)
a
```

```
DataFrame[model: string, mpg: double, cyl: int, disp: double, hp: int, drat: double, wt: double, qsec: double, vs: int, am: int, gear: int, carb: int]
```

```
top_five_mpg = a.orderBy(desc('mpg')).limit(5)
top_five_mpg.show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      model      |mpg|cyl|disp| hp|drat|  wt|  qsec|  vs| am|gear|carb|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Toyota Corolla |33.9| 4|71.1| 65|4.22|1.835| 19.9| 1| 1| 4| 1|
| Fiat 128 |32.4| 4|78.7| 66|4.08| 2.2|19.47| 1| 1| 4| 1|
| Honda Civic |30.4| 4|75.7| 52|4.93|1.615|18.52| 1| 1| 4| 2|
| Lotus Europa |30.4| 4|95.1|113|3.77|1.513| 16.9| 1| 1| 5| 2|
| Fiat X1-9 |27.3| 4|79.0| 66|4.08|1.935| 18.9| 1| 1| 4| 1|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

```
mpg_less_than_18 = a.filter(a.mpg < 18)
mpg_less_than_18.show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      model      |mpg|cyl|disp| hp|drat|  wt|  qsec|  vs| am|gear|carb|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Duster 360 |14.3| 8|360.0|245|3.21| 3.57|15.84| 0| 0| 3| 4|
| Merc 280C |17.8| 6|167.6|123|3.92| 3.44| 18.9| 1| 0| 4| 4|
| Merc 450SE |16.4| 8|275.8|180|3.07| 4.07| 17.4| 0| 0| 3| 3|
| Merc 450SL |17.3| 8|275.8|180|3.07| 3.73| 17.6| 0| 0| 3| 3|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18.0	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.25	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.0	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Dodge Challenger	15.5	8	318.0	150	2.76	3.52	16.87	0	0	3	2
AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.3	0	0	3	2
Camaro Z28	13.3	8	350.0	245	3.73	3.84	15.41	0	0	3	4
Ford Pantera L	15.8	8	351.0	264	4.22	3.17	14.5	0	1	5	4
Maserati Bora	15.0	8	301.0	335	3.54	3.57	14.6	0	1	5	8

```
df_with_wtTon = a.withColumn("wtTon", col("wt") * 0.45)
df_with_wtTon.show()
```

model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb	wtTon
Mazda RX4	21.0	6	160.0	110	3.9	2.62	16.46	0	1	4	4	1.179
Mazda RX4 Wag	21.0	6	160.0	110	3.9	2.875	17.02	0	1	4	4	1.29375
Datsun 710	22.8	4	108.0	93	3.85	2.32	18.61	1	1	4	1	1.044
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1	1.44675
Hornet Sportabout	18.7	8	360.0	175	3.15	3.44	17.02	0	0	3	2	1.548
Valiant	18.1	6	225.0	105	2.76	3.46	20.22	1	0	3	1	1.557
Duster 360	14.3	8	360.0	245	3.21	3.57	15.84	0	0	3	4	1.6065
Merc 240D	24.4	4	146.7	62	3.69	3.19	20.0	1	0	4	2	1.4355
Merc 230	22.8	4	140.8	95	3.92	3.15	22.9	1	0	4	2	1.4175
Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4	1.548
Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4	1.548
Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3	1.8315000000000001
Merc 450SL	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3	1.6785
Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18.0	0	0	3	3	1.7009999999999998
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.25	17.98	0	0	3	4	2.3625000000000003
Lincoln Continental	10.4	8	460.0	215	3.0	5.424	17.82	0	0	3	4	2.4408000000000003
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4	2.40525
Fiat 128	32.4	4	78.7	66	4.08	2.2	19.47	1	1	4	1	0.9900000000000001
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2	0.72675
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.9	1	1	4	1	0.82575

only showing top 20 rows

```
a.createOrReplaceTempView("mtcars")
sql_query_result = spark.sql("SELECT gear, cyl FROM mtcars WHERE cyl BETWEEN 4 AND 9")
sql_query_result.show()
```

gear	cyl
4	6
4	6
4	4
3	6
3	8
3	6
3	8
4	4
4	4
4	6
4	6
3	8
3	8
3	8
3	8
3	8
4	4
4	4
4	4

only showing top 20 rows

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